OPTIMIZING METRO PERFORMANCE

DATA-DRIVEN STRATEGIES TO IMPROVE RELIABILITY, SAFETY AND RESOURCE ALLOCATION FOR WMATA (METRO)

Last Updated

1[illegible]2025

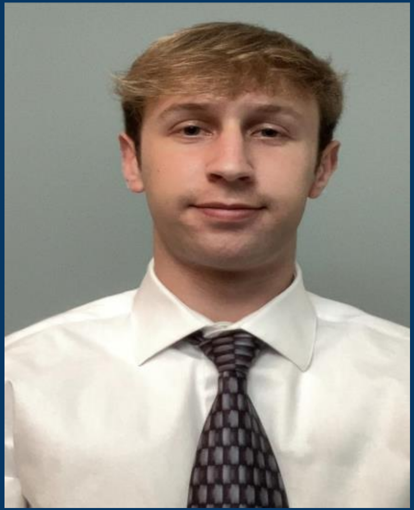
MEET THE TEAM

Data Science and Analytics

Vimbai Chivaura

Data Engineering and Infrastructure

Michael Mironenko



Transit Consulting and Strategy

Bokani Chuma

BACKGROUND

- **Complex, Multi-Jurisdiction Network:** WMATA's rail and bus operations span across multiple regions, each with its own demand patterns and operational constraints.
- **Silo Data Sources:** Performance logs, station data, bus routes, ridership counts, safety reports, and calendar data often sit in separate systems, limiting cross-system visibility.
- **Primary Users:** Operations managers, service planners, safety teams, and reliability analysts who need a unified view of how ridership, service patterns, and incidents interact.
- **MCC's Unique Approach:** This project integrates all key datasets into a single analysis-ready structure, enabling deeper diagnostics instead of surface-level reporting.
- **Why It Matters:** By consolidating the data, the analysis can uncover WMATA-specific inefficiencies, risk patterns, and operational opportunities that generic transit reports often miss.

INTRODUCTION

To anchor this work, the following mission statement defines the purpose of the analysis, while the mission objectives outline the key focus areas that will drive insight generation and support WMATA’s broader reliability and safety goals.

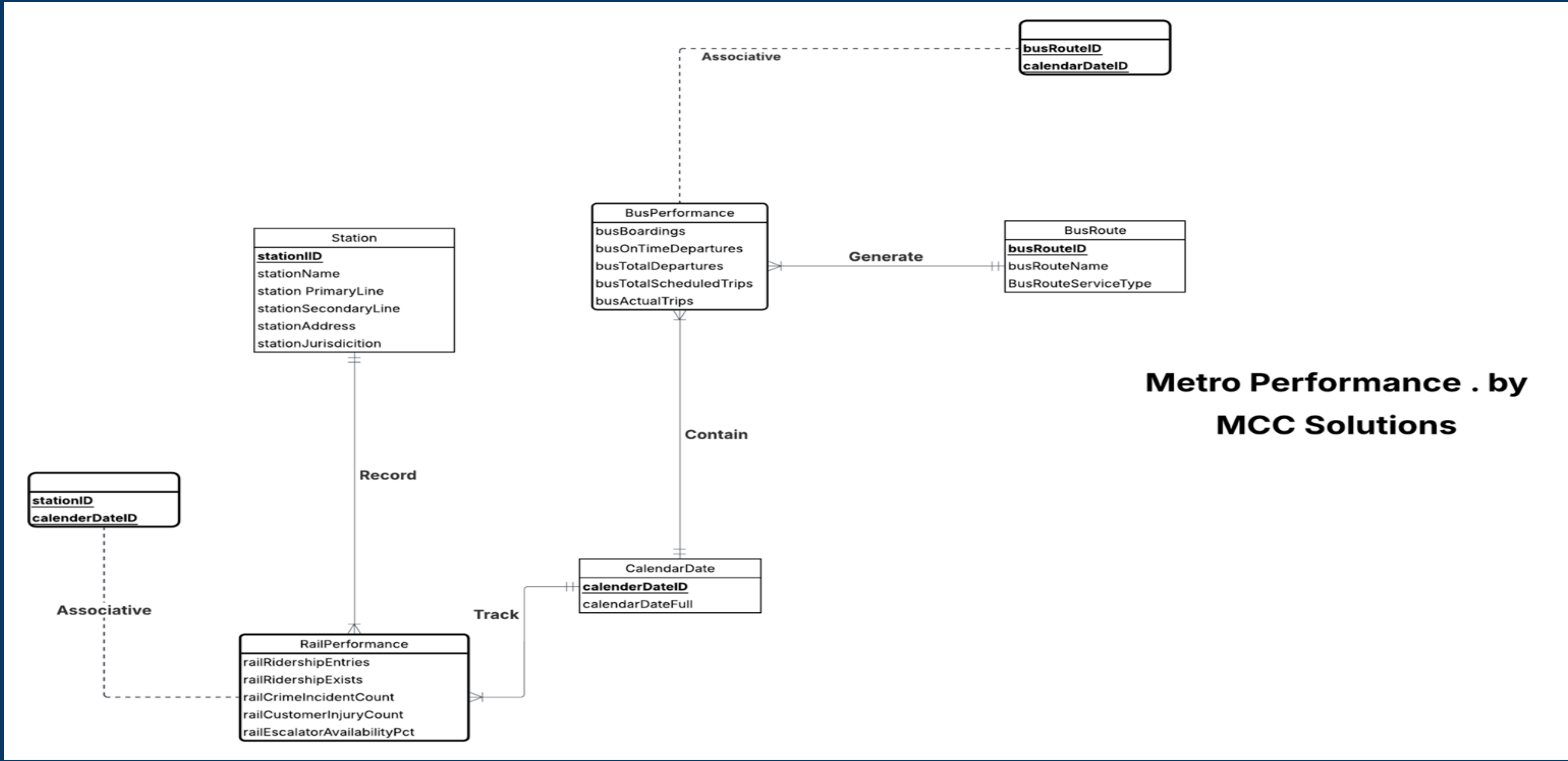
Mission Statement

To provide Washington Metropolitan Area Transit Authority (WMATA) with a comprehensive analysis that breaks down historical performance, ridership, and safety data. Our goal is to assist with data-driven decision-making to improve service reliability, enhance customer safety, and optimize resource allocation across the Metrorail and Metrobus systems.

Mission Objectives

1. **Analyze Ridership Trends:** Track and compare data for the period 2022-2024 relating to ridership patterns for each Metrorail station and Metrobus route, identifying high-growth and low-performing areas.
2. **Correlate service performance with reliability:** Evaluate operational metrics such as on-time departures to determine how reliability affects customer experience and ridership.
3. **Enhance safety and security insights:** Identify and map safety incidents by Metrorail station and line.
4. **Support Performance Ranking and Resource Allocation:** Enable ranking of stations and routes by ridership, reliability, and safety metrics to inform planning, maintenance prioritization, and service redesign initiatives.

CONCEPTUAL DATABASE DESIGN - ERD



LOGICAL DATABASE DESIGN: RELATIONAL SCHEMA

Relational Schema

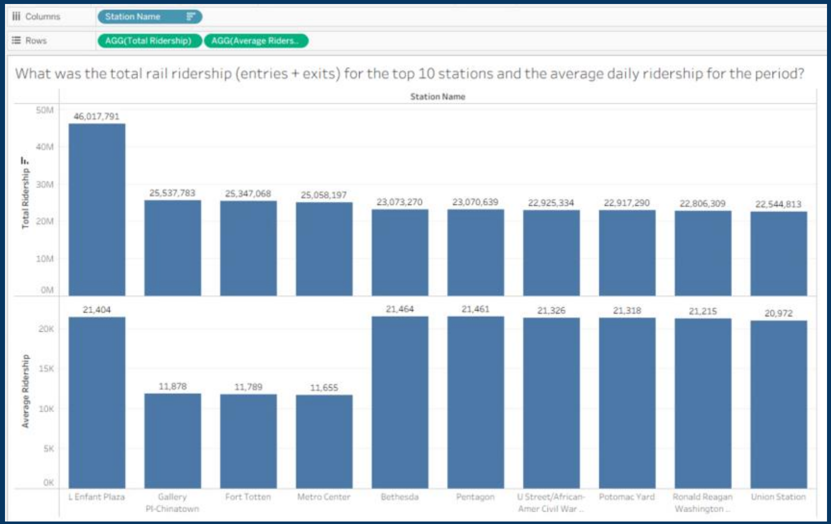
- Station (**stationID**, stationName, stationPrimaryLine, stationSecondaryLine, stationAddress, stationJurisdiction)
- BusRoute (**busRouteID**, busRouteName, busServiceType)
- CalendarDate (**calendardateID**, calendarFullDate)
- RailPerformance (***stationID***, ***calenderdateID***,ailRidershipEntries, railRidershipExits, railCrimeIncidentCount, railCustomerInjuryCount, railEscalatorAvailabilityPct)
- BusPerformance (***busRouteID***, ***dateID***, busBoardings, busOnTimeDepartures, busTotalDepartures, busTotalScheduledTrips, busActualTrips)

PHYSICAL DATABASE DESIGN

```
CREATE TABLE [Metro.RailPerformance] (
        calendarDateID DATE NOT NULL,
        stationID CHAR (5) NOT N
        railRidershipEntries INTEG
        railRidershipExits INTEGE
        railCrimeIncidentCount INT
        railCustomerInjuryCount INTEGER,
        railEscalatorAvailabilityPct DECIMAL (5,2),
        CONSTRAINT pk_RailPerformance_stationID_calendarDateID PRIMARY KEY (stationID, calendardateID),
        CONSTRAINT fk_RailPerformance_Station FOREIGN KEY (stationID)
                REFERENCES [Metro.Station] (stationID)
         ON DELETE NO ACTION ON UPDATE CASCADE,
        CONSTRAINT fk_RailPerformance_calendarDate FOREIGN KEY (calendardateID)
                REFERENCES [Metro.Calendardate] (calendardateID)
                ON DELETE CASCADE ON UPDATE CASCADE);
```

	Calendar Date ID	Station ID	Rail Ridership Entries	Rail Ridership Exits	Rail Crime Incident Count	Rail Customer Injury Count	Rail Escalator Availability Pct
1	2022-01-01	A01	7076	6745	0	0	92.68
2	2022-01-01	A02	2048	2056	0	0	92.75
3	2022-01-01	A03	1794	1803	0	0	92.79
4	2022-01-01	A04	616	601	0	0	92.17
5	2022-01-01	A05	939	915	0	0	94.03
6	2022-01-01	A06	709	712	0	0	93.57
7	2022-01-01	A07	650	657	0	0	93.48

Business Transaction 1



What was the total rail ridership (entries + exits) for each station and the average daily ridership for the period?

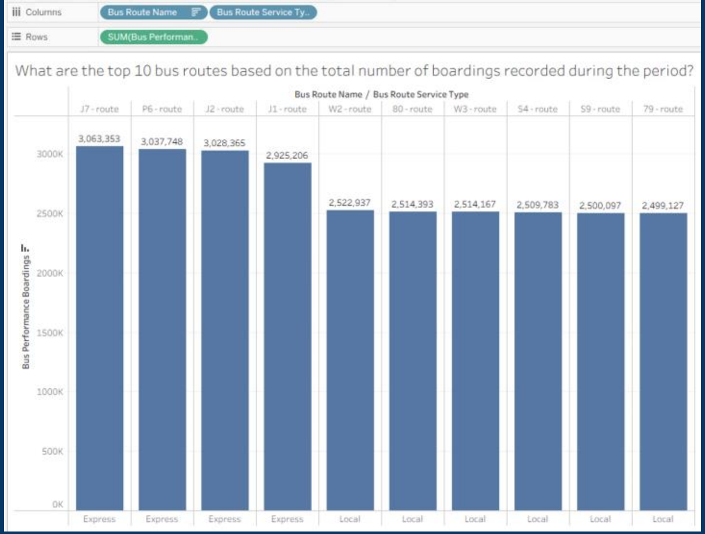
```
SELECT s.stationName AS 'Station Name' ,
SUM(p.railRidershipEntries + p.railRidershipExits)
AS 'Total Ridership',
AVG(p.railRidershipEntries + p.railRidershipExits)
AS 'Average Ridership'
FROM [Metro.RailPerformance] p JOIN
[Metro.Station] s
    ON p.stationID = s.stationID
GROUP BY s.stationName
ORDER BY [Total Ridership] DESC;
```

	Station Name	Total Ridership	Average Ridership
1	L Enfant Plaza	46017791	21403
2	Gallery Pl-Chinatown	25537783	11878
3	Fort Totten	25347068	11789
4	Metro Center	25058197	11654
5	Bethesda	23073270	21463
6	Pentagon	23070639	21461
7	U Street/African-Amer Civil War Memorial/Cardozo	22925334	21325
8	Potomac Yard	22917290	21318
9	Ronald Reagan Washington National Airport	22806309	21215
10	Union Station	22544813	20971

Business Transaction 2

What are the top 10 bus routes based on the total number of boardings recorded during the period?

```
SELECT TOP 10 r.busRouteName AS 'Bus Route',
r.busRouteServiceType AS 'Route Service Type',
SUM(b.busPerformanceBoardings) AS 'Total
Boardings'
FROM  [Metro.BusPerformance] b JOIN
[Metro.BusRoute] r
    ON b.busRouteID = r.busRouteID
GROUP BY r.busRouteID, r.busRouteName,
r.busRouteServiceType
ORDER BY [Total Boardings] DESC;
```

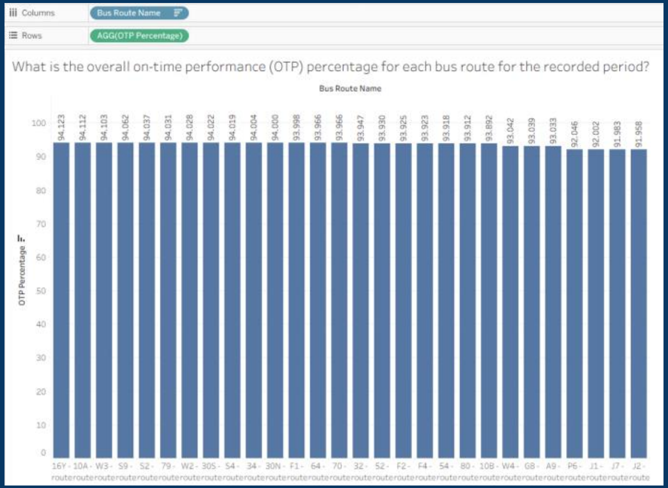


	Bus Route	Route Service Type	Total Boardings
1	J7 - route	Express	3063353
2	P6 - route	Express	3037748
3	J2 - route	Express	3028365
4	J1 - route	Express	2925206
5	W2 - route	Local	2522937
6	80 - route	Local	2514393
7	W3 - route	Local	2514167
8	S4 - route	Local	2509783
9	S9 - route	Local	2500097
10	79 - route	Local	2499127

Business Transaction 3

What is the overall on-time performance (OTP) percentage for each bus route for the recorded period?

```
SELECT r.busRouteName AS 'Bus Route',
(SUM(CAST(b.busPerformanceTimeDepartures AS
DECIMAL(18,6)))
    / NULLIF(SUM(CAST(b.busPerformanceTotalDepartures AS
DECIMAL(18,6))),0)) * 100
      AS 'OTP Percentage'
FROM [Metro.BusRoute] r JOIN [Metro.BusPerformance] b
    ON b.busRouteID = r.busRouteID
GROUP BY r.busRouteID, r.busRouteName
ORDER BY [Bus Route],[OTP Percentage] DESC;
```



	Bus Route	OTP Percentage
1	16Y - route	94.123400
2	10A - route	94.111700
3	W3 - route	94.103300
4	S9 - route	94.062300
5	S2 - route	94.036800
6	79 - route	94.031100
7	W2 - route	94.027600
8	30S - route	94.021800
9	S4 - route	94.018900
10	34 - route	94.003900
11	30N - route	93.999800
12	F1 - route	93.997500
13	64 - route	93.966100
14	70 - route	93.966100
15	32 - route	93.946600
16	52 - route	93.930000
17	F2 - route	93.925100
18	F4 - route	93.922900
19	54 - route	93.918000
20	80 - route	93.912100
21	10B - route	93.891900
22	W4 - route	93.041500
23	G8 - route	93.039300
24	A9 - route	93.033000
25	P6 - route	92.046400
26	J1 - route	92.001500
27	J7 - route	91.983000
28	J2 - route	91.958000

Business Transaction 4

What is the correlation between top 10 stations with highest incidents against stations highest?

```
WITH IncidentRank AS (
   SELECT TOP 10 s.stationID,s.stationName AS 'Station Name',s.stationJurisdiction AS 'Station Jurisdiction',
   SUM(r.railCrimeIncidentCount + r.railCustomerInjuryCount) AS 'Total Incidents',
   RANK() OVER (ORDER BY SUM(r.railCrimeIncidentCount + r.railCustomerInjuryCount) DESC) AS 'Incident
Rank'
   FROM [Metro.Station] s
   JOIN [Metro.RailPerformance] r ON s.stationID = r.stationID
   GROUP BY s.stationID, s.stationName, s.stationJurisdiction),RidershipRank AS (
                                             SELECT  s.stationID,s.stationName,
      SUM(r.railRidershipEntries + r.railRidershipExits) AS 'Total Ridership',
      RANK() OVER (ORDER BY SUM(r.railRidershipEntries + r.railRidershipExits) DESC) AS 'Ridership Rank'
                                             FROM [Metro.Station] s
                                             JOIN [Metro.RailPerformance] r  ON s.stationID = r.stationID
      GROUP BY s.stationID, s.stationName)
                                                               SELECT [Station Name],[Station Jurisdiction],[Total Incidents],[Incident
Rank],[Total Ridership], [Ridership Rank], ([Ridership Rank] - [Incident Rank]) AS 'Rank Difference'
                                                               FROM IncidentRank i JOIN RidershipRank r ON i.stationID =
r.stationID
ORDER BY [Incident Rank];
```

Business Transaction 4

	Station Name	Station Jurisdiction	Total Incidents	Incident Rank	Total Ridership	Ridership Rank	Rank Difference
1	L Enfant Plaza	Washington DC	113	1	23112043	2	1
2	U Street/African-Amer Civil War Memorial/Cardozo	Washington DC	103	2	22925334	6	4
3	Metro Center	Washington DC	102	3	22795384	10	7
4	Gallery Pl-Chinatown	Washington DC	102	3	23280694	1	-2
5	Rosslyn	Arlington VA	102	3	22385274	12	9
6	Ronald Reagan Washington National Airport	Arlington VA	101	6	22806309	9	3
7	Union Station	Washington DC	101	6	22544813	11	5
8	Bethesda	Bethesda MD	100	8	23073270	4	-4
9	Pentagon	Arlington VA	100	8	23070639	5	-3
10	Potomac Yard	Alexandria VA	99	10	22917290	7	-3

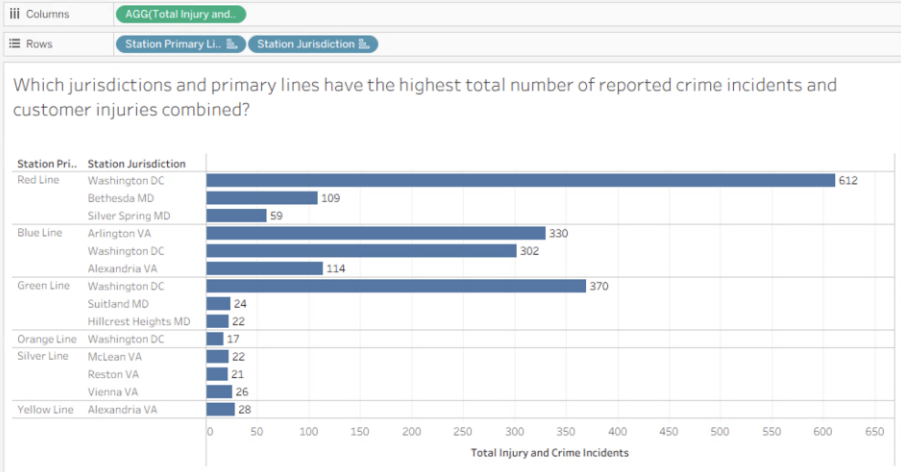
Business Transaction 5

Which jurisdiction and primary line has the highest total number of reported crime incidents and customer injuries combined?

```
SELECT TOP 10 s.stationJurisdiction AS 'Station
Jurisdiction',s.stationPrimaryLine AS 'Station PrimaryLine' ,
SUM(r.railCrimeIncidentCount) AS TotalReportedCrimes,SUM
(r.railCustomerInjuryCount) AS TotalCustomerInjury,
SUM(r.railCrimeIncidentCount +r.railCustomerInjuryCount) AS 'Total Safety
Incidents'
FROM [Metro.Station] s JOIN [Metro.RailPerformance] r
    ON  s.stationID = r.stationID
GROUP BY s.stationJurisdiction , s.stationPrimaryLine
ORDER BY [Total Safety Incidents] DESC;
```

Business Transaction 5

Which jurisdiction and primary line has the highest total number of reported crime incidents and customer injuries combined?



	Station Jurisdiction	Station PrimaryLine	TotalReportedCrimes	TotalCustomerInjury	Total Safety Incidents
1	Washington DC	Red Line	348	264	612
2	Washington DC	Green Line	201	169	370
3	Arlington VA	Blue Line	179	151	330
4	Washington DC	Blue Line	161	141	302
5	Alexandria VA	Blue Line	63	51	114
6	Bethesda MD	Red Line	61	48	109
7	Silver Spring MD	Red Line	29	30	59
8	Alexandria VA	Yellow Line	19	9	28
9	Vienna VA	Silver Line	13	13	26
10	Suitland MD	Green Line	15	9	24

Business Transaction 6

Which rail stations had the highest total number of crime incidents and customer injuries, and what was the total ridership for those stations?

```
SELECT TOP 10 s.stationName AS 'Station Name', SUM(p.railCrimeIncidentCount)  AS 'Count of Crime Incidents', SUM(p.railCustomerInjuryCount) AS 'Count of Customer Injuries',

 SUM(p.railRidershipEntries + p.railRidershipExits) AS 'Total Ridership'

FROM [Metro.Station] s JOIN [Metro.RailPerformance] p

    ON s.stationID = p.stationID

GROUP BY s.stationName

ORDER BY [Count of Crime Incidents] DESC, [Count of Customer Injuries] DESC;
```

BUSINESS TRANSACTION 6

Which rail stations had the highest total number of crime incidents and customer injuries, and what was the total ridership for those stations?

	Station Name	Count of Crime Incidents	Count of Customer Injuries	Total Ridership
1	L Enfant Plaza	104	103	46017791
2	Metro Center	69	42	25058197
3	Gallery Pl-Chinatown	67	49	25537783
4	Ronald Reagan Washington National Airport	60	41	22806309
5	Rosslyn	59	43	22385274
6	Bethesda	57	43	23073270
7	Fort Totten	57	43	25347068
8	Potomac Yard	54	45	22917290
9	Union Station	51	50	22544813
10	Pentagon	50	50	23070639

DASHBOARD

What was the total rail ridership (entries + exits) for the top 10 stations and the average daily ridership for the period?
Station Name
Total Ridership
Average Riders..
46,017,791
25,537,783
25,347,068
25,058,197
23,073,270
23,070,639
22,925,334
22,917,290
22,806,309
22,544,813
21,404
11,878
11,789
11,655
21,464
21,461
21,326
21,318
21,215
20,972
L Enfant Pl..
Gallery Pl-..
Fort Totten
Metro Cen..
Bethesda
Pentagon
U Street/A..
Potomac Y..
Ronald Re..
Union Stat..
What are the top 10 bus routes based on the total number of boardings recorded during the period?
Bus Route Name / Bus Route Service Type
Bus Performance Boardings
3,063,353
3,037,748
3,028,365
2,925,206
2,522,937
2,514,393
2,514,167
2,509,783
2,500,097
2,499,127
Measure Names
Total Injury and Crime Incidents
Which jurisdictions and primary lines have the highest total number of reported crime incidents and customer injuries combined?
Station Pri..
Station Jurisdi..
Red Line
Washington DC 612
Bethesda MD 109
Silver Spring MD 59
Blue Line
Arlington VA 330
Washington DC 302
Alexandria VA 114
Green Line
Washington DC 370
Suitland MD 24
Hillcrest Heigh.. 22
Orange Line
Washington DC 17
Silver Line
McLean VA 22
Reston VA 21
Vienna VA 26
Yellow Line
Alexandria VA 28
Total Injury and C..
What is the overall on-time performance (OTP) percentage for each bus route for the recorded period?
Bus Route Name
OTP Percentage
94.123
94.103
94.037
94.028
94.019
94.000
93.966
93.947
93.925
93.918
93.892
93.039
92.046
91.983
Which rail stations had the highest total number of crime incidents and customer injuries, and what was the total ridership for those stations?
Station Name / Total Ridership
Total Injury and Crime Incidents
207
116
111
103
102
101
101
100
100
100

INSIGHTS

• **Ridership Concentrates at Specific High-Demand Stations**
Total entries and exits show that a small number of stations account for the majority of ridership. These locations experience the highest operational pressure and require more targeted management.

• **Safety Incidents Do Not Always Align With Ridership Levels**
Several stations record high crime and injury incidents even with moderate ridership. This indicates location-specific safety risks that are not merely a function of crowd size

• **Jurisdictional differences influence station performance patterns**, with some regions reporting higher safety incidents despite similar ridership volumes.Safety and ridership patterns differ across jurisdictions, revealing regional variations in operational risk.

RECOMMENDATIONS

- **Increase Service Frequency and Staff Support at High-Demand Stations**

Adjust headways, allocate platform personnel, and improve crowd-flow management at stations with sustained high ridership volumes.

- **Implement Targeted Safety Measures at Identified Incident Hotspots**

Deploy transit police, enhance lighting, and increase surveillance where crime and injury counts exceed expected levels relative to ridership.

RECOMMENDATIONS

- **Use Calendar-Based Trends to Improve Bus Reliability**

Reinforce scheduling and vehicle deployment on days/routes with recurring performance gaps, reducing missed or delayed trips.

- **Prioritize Maintenance for Low Escalator Availability Stations**

Focus escalator repairs and preventive maintenance at stations with high ridership but low availability percentages to maintain accessibility and flow


THANK YOU